Softtek Knowledge Management System

Case study: Softtek · Lifecycle, ontology, automated classification and semantic search · Project progress, October 2026



Softtek: Mexican IT services firm founded in 1982 in Monterrey

Type of organization

Private IT services and consulting company; ~15,000 employees, 30 offices; pioneer of the Nearshore model

Core activities

- Agile development and application modernization
- QA, DevOps and site reliability engineering
- SAP and Salesforce services
- Application, cloud and end-user support
- Digital consulting and cognitive automation

Source: Softtek Fast Facts 2024 and Wikipedia. Documents and results are illustrative.

KNOWLEDGE HANDLED: 5 DOCUMENT TYPES

Technical Documentation
API references, architecture, deployment guides

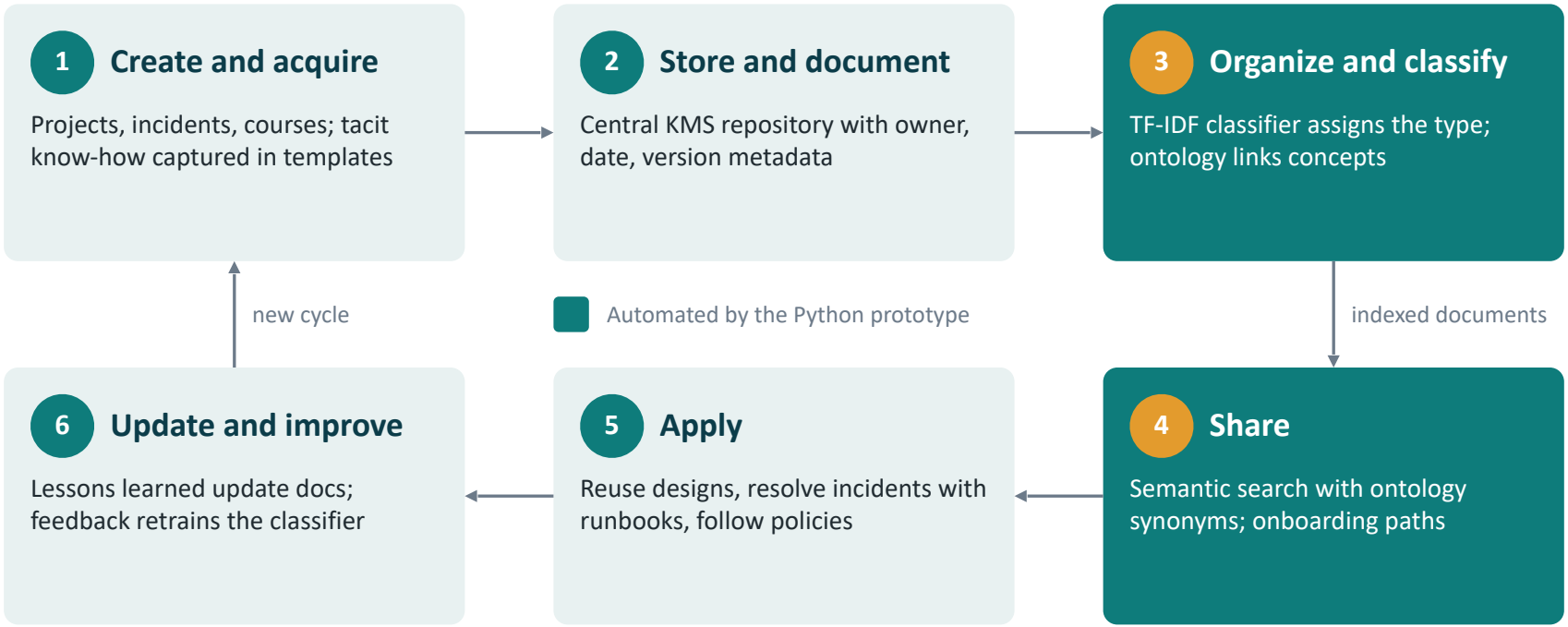
Policies & Procedures
Security, remote work, expenses, change control

Project Reports
Status reports, sprint reviews, lessons learned

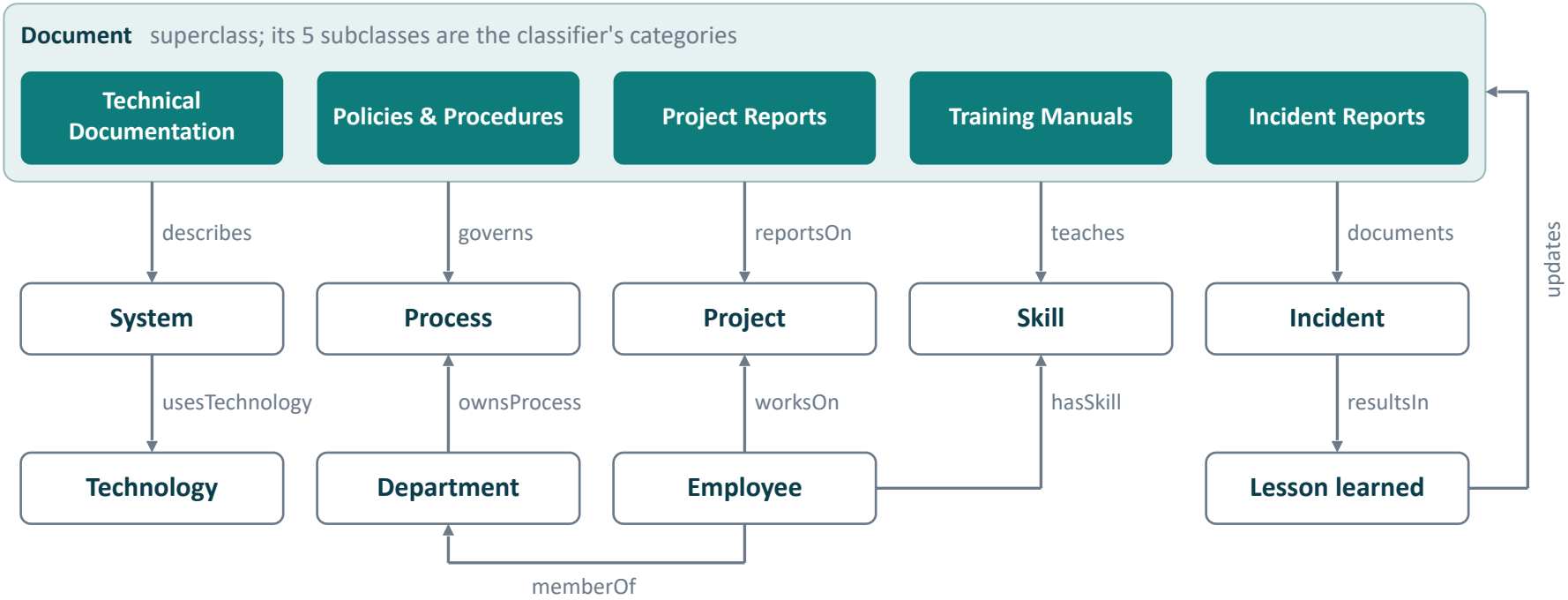
Training Manuals
Onboarding guides, courses, tutorials

Incident Reports
Post-mortems, root-cause analyses

Knowledge moves through a closed six-stage cycle



The ontology anchors each document type to a domain concept



OWL/SKOS file ontology/softtek_ontology.ttl · 13 domain classes, 5 document classes, 22 relationships, 372 triples

Typed relationships make the knowledge graph queryable

Property	From → To	Example
describes	Technical Doc → System	TD01 → Payments API
governs	Policy → Process	PP07 → Change management
reportsOn	Project Report → Project	PR10 → Mobile banking
teaches	Training Manual → Skill	TM03 → Kubernetes operation
documents	Incident Report → Incident	IR01 → Payments outage
references	Document → Document	IR01 → TD01
updates	Lesson learned → Document	Add perf. test → TD04, PP07



Competency question

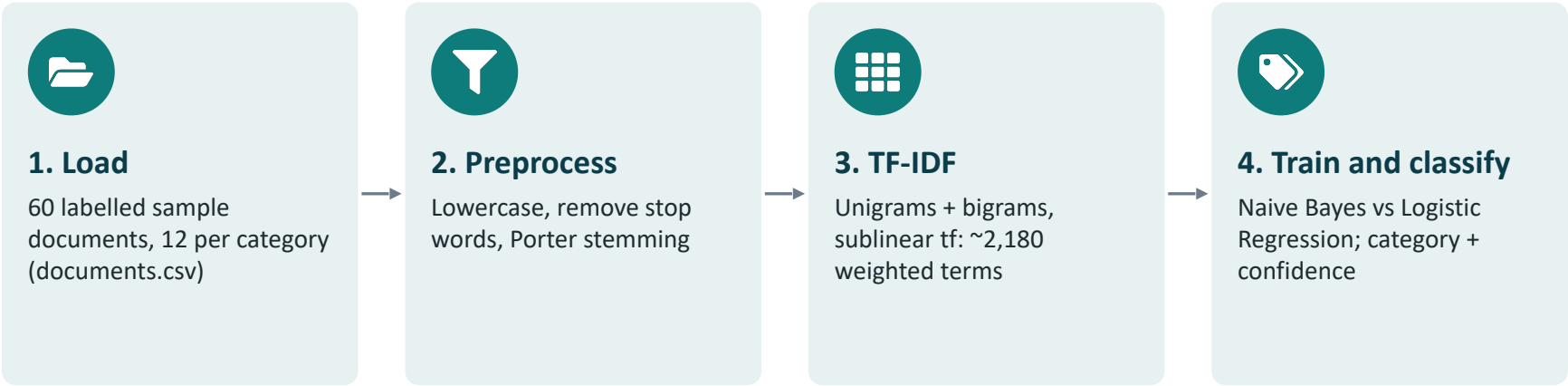
Which documents must be updated after incidents?

Payments outage
→ Logging and Monitoring Setup

Inventory DB degradation
→ Database Schema (TD04)
→ Change Management (PP07)

Answered by exercises/ex6_ontology_queries.py

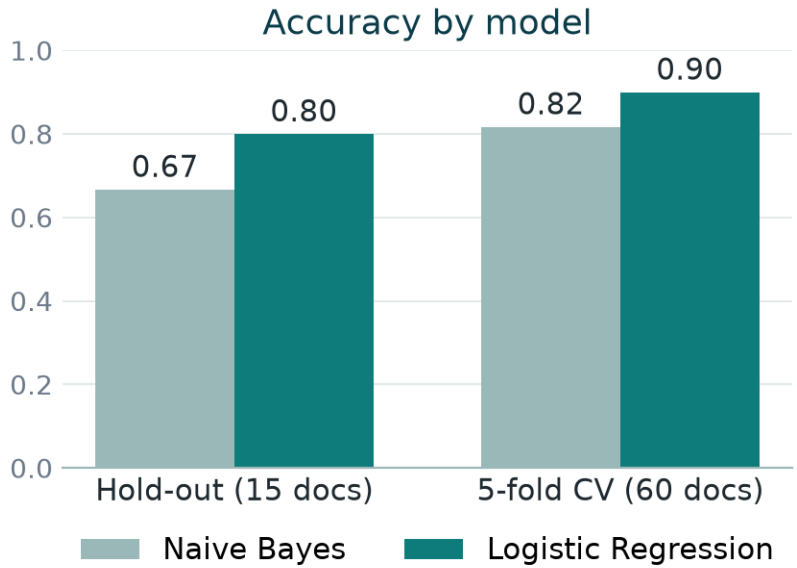
The classifier turns raw text into a document type in four steps

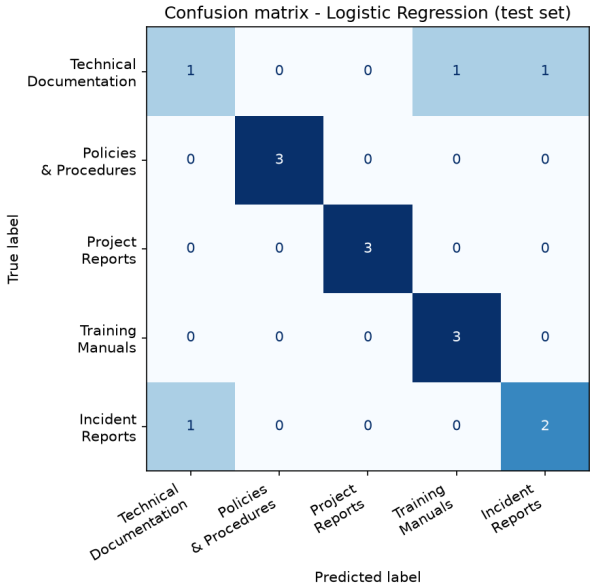


TF-IDF weight: $\text{tfidf}(t, d) = (1 + \ln \text{tf}_{t,d}) \times (\ln((1 + N) / (1 + \text{df}_t)) + 1)$, then each vector is L2-normalized

Run: `python src/classifier.py` | `python src/classifier.py --text "..."`

Logistic Regression classifies document types at 0.90 accuracy





Hold-out has 3 docs per class: one error moves accuracy by 0.07. Cross-validation is the reliable figure.

Errors only between Technical Documentation and Incident Reports, which share words like API, service, deploy.

Five of six unseen documents were classified correctly

Id	Text (start)	Predicted	Conf.	OK
N01	The checkout service was down for 25 minutes after a certificate expired…	Incident Reports	0.61	Yes
N02	Employees must lock their screen when leaving the desk…	Policies & Procedures	0.32	Yes
N03	The search service exposes a GET /v1/search endpoint…	Technical Documentation	0.49	Yes
N04	Module 3 of the course teaches how to write SQL queries…	Training Manuals	0.69	Yes
N05	The project is two weeks behind schedule; budget spent is 70%…	Project Reports	0.76	Yes
N06	New team members should complete the onboarding lessons on Docker…	Project Reports	0.34	No



N06 should be a Training Manual: "team" and "client project" are strong Project Report terms. Predictions below 0.40 confidence could go to a human reviewer.

Search ranks documents by cosine similarity to the query

1

Natural-language query

"what happened when the payment system went down"

2

Ontology expansion

SKOS altLabel "went down" adds "outage incident"

3

TF-IDF vector

Same fitted vectorizer as the documents

4

Cosine similarity

Score against all 60 document vectors

5

Top-k results

IR01 Payments API Outage ranked first (0.197)

cos(q, d) = (q · d) / (||q|| ||d||)

1 = same term distribution, 0 = no shared terms. Run: `python src/semantic_search.py --demo`

Stemming and ontology synonyms put the right document first

Query: "what happened when the payment system went down"

Plain TF-IDF (baseline)

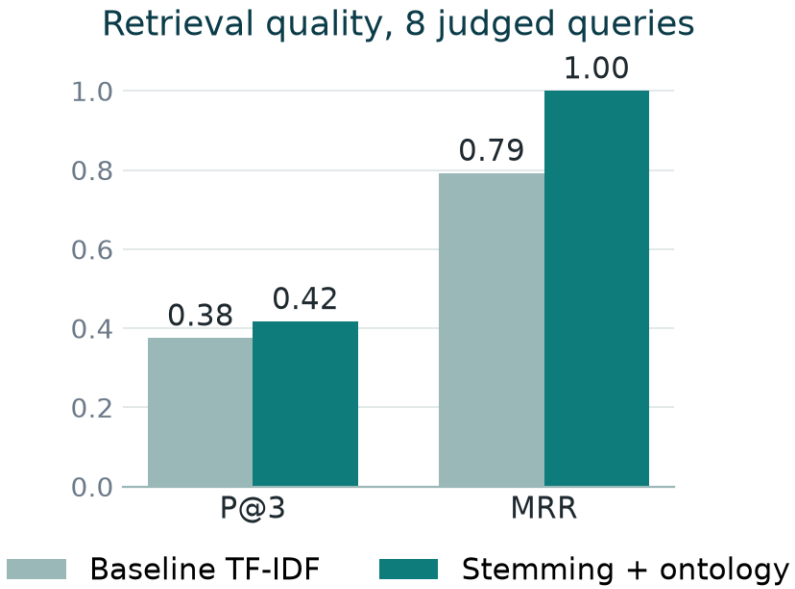
1. IR10 Cache Stampede post-mortem (0.103)
2. PR01 Retail Platform status report (0.098)

IR01 not found: it says "Payments" and "Outage"

Stemming + ontology expansion

1. IR01 Post-mortem: Payments API Outage (0.197)

2. IR11 Email Service Outage (0.195)
3. IR04 Data Center Network Failure (0.168)



MRR 1.00: a relevant document is ranked first for every test query.

Six exercises from Topics 1–6 built the prototype's pieces

1. Text preprocessing

Vocabulary 986 words → 704 stems (29% smaller)

2. TF-IDF by hand

Manual values equal scikit-learn exactly

3. Cosine similarity

Same-type docs 2–5× more similar than other types

4. Classification tests

4 representations × 3 models; best 0.90 CV accuracy

5. Retrieval evaluation

8 judged queries: MRR 0.79 → 1.00

6. Ontology queries

5 SPARQL competency questions on 372 triples

Basic techniques already cover the KMS core

TF-IDF + Logistic Regression classifies at 0.90; ontology-expanded search ranks a relevant document first for every test query

Next steps

- Grow the collection beyond 60 documents
- Send low-confidence predictions to human review
- Add sentence embeddings for meaning-based search
- Link new documents to the ontology automatically